WORKED SOLUTION

STEP III 1991, Question 13

Mechanics • Collision

STEP III 1991, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

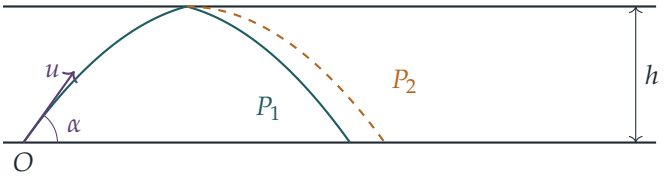
step.maths.org/questions/91-s3-q13

The key idea

The ceiling is smooth, so the horizontal velocity $u\cos\alpha$ never changes and every range is just $u\cos\alpha$ times a flight time. All the work is in the vertical motion, and the energy equation gives the vertical speed at the ceiling straight away. After an elastic impact the particle comes down exactly as it went up, played backwards; after an inelastic one it simply drops from rest.

Solution

Model both particles as point masses moving freely under gravity, with no air resistance. Write $u = \sqrt{8gh}$, $s = \sin\alpha$ and $c = \cos\alpha$. Take x horizontal and y vertically up from O.



Reaching the ceiling

The initial vertical velocity is us, and while rising

$$\dot{y}^2 = u^2s^2 - 2gy = 8gh\,s^2 - 2gy.$$

The particle reaches height h only if $\dot{y}^2 \geq 0$ there, that is $8ghs^2 \geq 2gh$, so $s^2 \geq \frac{1}{4}$. Since α is acute, $s > 0$ and this is $\sin\alpha \geq \frac{1}{2}$, i.e. $\alpha \geq \frac{1}{6}\pi$. (At $\alpha = \frac{1}{6}\pi$ exactly it just grazes the ceiling.) When it does arrive, its upward speed there is

$$w = \sqrt{8ghs^2 - 2gh} = \sqrt{2gh}\,\sqrt{4s^2 - 1}.$$

Range of P_1

On the way up, the vertical velocity falls steadily from us to w at rate g, so the time to the ceiling is

$$t_1 = \frac{us - w}{g}.$$

Ask yourself How long does P_1 take to come back down from the ceiling?

The impact is perfectly elastic and the ceiling is smooth, so the vertical velocity becomes w downwards and the horizontal velocity is unchanged. The vertical motion from the ceiling down is then the vertical motion on the way up run backwards in time (start at height h with speed w towards the floor, accelerate at g). So it arrives at the floor with vertical speed us, taking another time t_1. Directly: the vertical speed grows from w to $\sqrt{w^2 + 2gh} = us$ at rate g.

The total flight time is $2t_1$, so the range is

$$\begin{aligned} R_1 &= uc \cdot \frac{2(us - w)}{g} = \frac{2\sqrt{8gh}\,c}{g}\left(2\sqrt{2gh}\,s - \sqrt{2gh}\sqrt{4s^2 - 1}\right) \\ &= \frac{2\sqrt{8gh}\sqrt{2gh}}{g}\,c(2s - \sqrt{4s^2 - 1}) = 8h\cos\alpha\left(2\sin\alpha - \sqrt{4\sin^2\alpha - 1}\right), \end{aligned}$$

using $us = \sqrt{8gh}\,s = 2\sqrt{2gh}\,s$ and $\sqrt{8gh}\sqrt{2gh} = 4gh$. This is the required range.

Range of P_2 and the difference D

P_2 follows the same path up to the ceiling, taking time t_1. The impact is perfectly inelastic, so its vertical velocity becomes zero; the ceiling is smooth, so the horizontal velocity is still uc. It then falls a height h from rest, taking time $\sqrt{2h/g}$. Hence

$$R_2 = uc\left(t_1 + \sqrt{\tfrac{2h}{g}}\right), \qquad R_2 - R_1 = uc\left(\sqrt{\tfrac{2h}{g}} - t_1\right).$$

Now

$$t_1 = \frac{\sqrt{2gh}\,(2s - \sqrt{4s^2 - 1})}{g} = \sqrt{\frac{2h}{g}}\,(2s - \sqrt{4s^2 - 1})$$

and $uc\sqrt{2h/g} = \sqrt{8gh}\sqrt{2h/g}\,c = 4hc$, so

$$R_2 - R_1 = 4h\cos\alpha\left(1 - 2\sin\alpha + \sqrt{4\sin^2\alpha - 1}\right).$$

This is positive for $\frac{1}{6}\pi < \alpha < \frac{1}{2}\pi$: for $s > \frac{1}{2}$ we have $(2s-1)^2 = 4s^2 - 1 - 2(2s-1) < 4s^2 - 1$, so $2s - 1 < \sqrt{4s^2 - 1}$. So P_2 always lands further away (the elastic bounce throws P_1 down faster), and

$$D = 4h\cos\alpha\left(1 - 2\sin\alpha + \sqrt{4\sin^2\alpha - 1}\right).$$

Maximising D

Let $f(\alpha) = D/(4h) = c(1-2s) + cr$, where $r = \sqrt{4s^2-1}$, on $\frac{1}{6}\pi \le \alpha < \frac{1}{2}\pi$. Note $f(\frac{1}{6}\pi) = 0$, $f \to 0$ as $\alpha \to \frac{1}{2}\pi$ (since $c \to 0$), and $f > 0$ in between. Differentiating, with $\frac{dr}{d\alpha} = \frac{4sc}{r}$,

$$f'(\alpha) = -s(1-2s) - 2c^2 - sr + \frac{4sc^2}{r}.$$

Putting $c^2 = 1 - s^2$ and multiplying by $r > 0$ (for $\alpha > \frac{1}{6}\pi$):

$$\begin{aligned} rf'(\alpha) &= (4s^2 - s - 2)\,r - sr^2 + 4s(1-s^2) \\ &= (4s^2 - s - 2)\,r - s(8s^2 - 5). \end{aligned}$$

At $\alpha = \frac{1}{4}\pi$, $s = \frac{1}{\sqrt{2}}$ and $r = 1$, giving $(2 - \frac{1}{\sqrt{2}} - 2) - \frac{1}{\sqrt{2}}(4-5) = 0$. So $\alpha = \frac{1}{4}\pi$ is a stationary point.

Ask yourself How do we know it's the maximum, not some other stationary point?

At any stationary point, $(4s^2 - s - 2)r = s(8s^2 - 5)$. Squaring and using $r^2 = 4s^2 - 1$, expanding and factorising gives

$$(4s^2 - s - 2)^2(4s^2 - 1) - s^2(8s^2 - 5)^2 = -2(2s^2 - 1)(8s^3 - s^2 - 2s - 2),$$

so every stationary point has $s^2 = \frac{1}{2}$ or $q(s) = 8s^3 - s^2 - 2s - 2 = 0$. On $[\frac{1}{2}, 1]$, $q'(s) = 24s^2 - 2s - 2 > 0$, so q is increasing; since $q(0.8) = -0.144 < 0 < 0.0985 = q(0.82)$, its only root there lies in $(0.8, 0.82)$. But for such s,

$$4s^2 - s - 2 < 4(0.82)^2 - 0.8 - 2 < 0, \qquad 8s^2 - 5 > 8(0.8)^2 - 5 > 0,$$

so the left side $(4s^2 - s - 2)r$ is negative and the right side $s(8s^2 - 5)$ is positive: this root came from squaring and is not a stationary point. Hence $\alpha = \frac{1}{4}\pi$ is the only stationary point in $(\frac{1}{6}\pi, \frac{1}{2}\pi)$. Since f is continuous, positive inside the interval and zero (or tending to zero) at both ends, it has a maximum inside, which must be at this stationary point. So D is greatest when $\alpha = \frac{1}{4}\pi$, with

$$D_{\max} = 4h \cdot \tfrac{1}{\sqrt{2}}(1 - \sqrt{2} + 1) = 4h(\sqrt{2} - 1).$$

Checks

- At $\alpha = \frac{1}{6}\pi$ the particle just touches the ceiling, so the two particles should agree: indeed $D = 0$, and $R_1 = 8h \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}\,h$ is the usual range $u^2 \sin 2\alpha / g$.
- With $h = 1.7\,\text{m}$, stepping both motions numerically at $\alpha = 1$ gives $R_1 \approx 2.420\,\text{m}$ and $D \approx 2.464\,\text{m}$, matching the formulae. A fine search over α puts the maximum of $D/(4h)$ at $\alpha = 0.78540 \approx \frac{1}{4}\pi$ with value $0.41421 \approx \sqrt{2} - 1$.

Common slips

- Solving $h = ust - \frac{1}{2}gt^2$ for t_1 and taking the larger root, which is the time the particle would pass height h on the way down if there were no ceiling.
- Writing $D = R_1 - R_2$, which is negative: the inelastic particle lands further away.
- Stopping at $f'(\frac{1}{4}\pi) = 0$ without showing that this stationary point is a maximum.

Beyond STEP

The time-reversal argument for P_1 is a first glimpse of a deep fact: Newton's laws with conservative forces look the same when time runs backwards, and a perfectly elastic impact preserves that symmetry, while an inelastic one (which loses energy) breaks it. Coefficients of restitution between 0 and 1 would interpolate between the two ranges found here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and energy conservation in this spirit.

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